Arun Vignesh Malarkkan

malarkkanarunvignesh@gmail.com (+1) 480-255-2954 http://arunvignesh28.github.io/

About Me

- Investigating reliability and generalization of LLM reasoning, combining causal inference and mechanistic interpretability, focusing on invariance discovery, spurious correlation mitigation, and robust evaluation.

Education

Aug 2023 – Present *(Expected: Fall 2026)*	**Ph.D., Computer Science**, Arizona State University, Tempe, AZ. Advisor: Dr. Yanjie Fu. Knowledge Discovery and Data Mining Lab. CGPA: 3.9/4.0.
May 2020	**Master of Science, Computer Science**, Arizona State University, Tempe, AZ. CGPA: 3.8/4.0.

Technical Skills

Core Expertise	LLM Reasoning & Reliability, Recommendation Systems, Causal Inference & Causal Representation Learning, Mechanistic Interpretability, Post-training & Fine-tuning, Retrieval-Augmented Generation (RAG), Data-Centric AI.
ML & Systems	PyTorch, TensorFlow, Hugging Face Transformers, Distributed Training, GPU/CUDA Programming.
LLM & Generative AI	PEFT (LoRA, QLoRA), LangChain, LlamaIndex, OpenAI API, vLLM.
Interpretability & Analysis	TransformerLens, Activation Patching, Probing, PyTorch Hooks.
Programming	Python, Java, TypeScript.
Data & Infrastructure	AWS, GCP, BigQuery, MySQL, Apache Spark (SparkML), Docker.
General Libraries	Scikit-learn, Pandas, NumPy, Flask, AutoGluon.

Research Publications

First-Author Publications

1. **A. V. Malarkkan**, U. Byahut, M. R. Choudhury, Y. Charde, V. Gupta, and Y. Fu, "Invariant reasoning directions in latent trajectories of language models," in *ICML Workshop on Mechanistic Interpretability*, 2026.
2. **A. V. Malarkkan**, X. Wang, K. Liu, D. Zhang, and Y. Fu, "Causally-guided diffusion for stable feature selection," 2026, arXiv preprint. Under review, KDD.
3. **A. V. Malarkkan**, W. Ying, and Y. Fu, *CAFE: Causally-guided automated feature engineering with multi-agent reinforcement learning*, arXiv preprint. Under review, NEURIPS, 2026.
4. **A. V. Malarkkan** et al., *FinRule-Bench: A benchmark for joint reasoning of llms over financial tables and principles*, arXiv preprint. Under review, EMNLP, 2026.
5. **A. V. Malarkkan** et al., *Latent reasoning does not lack signal, it lacks invariance: Subspace-constrained refinement for robust latent reasoning*, arXiv preprint. Under review, NeurIPS, 2026.
6. **A. V. Malarkkan**, H. Bai, A. Kaushik, and Y. Fu, "DELTA: Variational disentangled learning for privacy-preserving data reprogramming," in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2025.
7. **A. V. Malarkkan**, H. Bai, D. Wang, and Y. Fu, "Causal graph profiling via structural divergence for robust anomaly detection in cyber-physical systems," in *Proceedings of the IEEE International Conference on Big Data*, 2025.

8. **A. V. Malarkkan**, D. Wang, H. Bai, and Y. Fu, "Incremental causal graph learning for online cyberattack detection in cyber-physical infrastructures," *IEEE Transactions on Big Data*, 2025.
9. **A. V. Malarkkan**, D. Wang, and Y. Fu, "Multi-view causal graph fusion based anomaly detection in cyber-physical infrastructures," in *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2024.

Co-Authored Publications

1. H. Cao et al., *Not all correct gradients should be trusted: Shortcut-aware reasoning via directional detection and correction*, arXiv preprint. Under review, NeurIPS, 2026.
2. H. Cao et al., *Robust data reshaping via flat and consensus-seeking diffusion*, arXiv preprint. Under review, NeurIPS, 2026.
3. X. Wang, K. Liu, **A. V. Malarkkan**, and Y. Fu, *Evolving demonstration optimization for chain-of-thought feature transformation*, arXiv preprint. Under review, KDD, 2026.
4. W. Ying et al., *Distribution shift aware neural tabular learning*, arXiv preprint. Under review, ICML, 2026.
5. W. Ying et al., *A survey on data-centric ai: Tabular learning from reinforcement learning and generative ai*, arXiv preprint., 2025.

Professional Experience

May 2026 – Present

Research Intern : LLM Post-Training, The Home Depot.

- Post-training 4B and 8B small language models for product compatibility reasoning, enabling catalog-scale verification of whether products can be correctly used, installed, or purchased together.
- Developing a novel preference optimization framework that isolates stable compatibility-relevant learning signal from unstable catalog-specific variation during post-training, improving robustness to noisy metadata, paraphrased queries, and product taxonomy drift.
- Building trajectory- and prediction-level evaluation protocols that stress-test compatibility reasoning across semantically equivalent product-pair queries, measuring accuracy, invariance, calibration, and consistency under distribution shift.

Jun 2025 – Present

Doctoral Researcher, DOW Chemicals (National Academy of Engineering Grant).

- Leading research on incorporating LLM-guided knowledge priors into neuro-symbolic, causally-aware multi-agent reinforcement learning frameworks to encode domain constraints and guide structured reasoning in materials discovery.
- Developing frameworks that integrate causal structure with sequential decision-making to improve robustness under distribution shifts in industrial simulation environments.
- Designing interpretable and scalable models for material optimization under uncertainty (details subjected to confidentiality).

Jan 2023 – Jun 2023

Senior Software Engineer, Fidelity Investments.

- Led development of Monte Carlo-based financial projection systems for retirement planning at scale.
- Built probabilistic simulation frameworks for uncertainty-aware decision modeling in long-term financial forecasting.
- Developed ensemble-based outlier detection models to identify key drivers affecting projected savings outcomes.

Professional Experience (continued)

Jun 2021 – Jan 2023 **Software Development Engineer,** Amazon — Alexa AI.

- Developed and deployed purchase likelihood modeling for Alexa recommendation systems, improving voice-driven purchases by $+8\%$.
- Built end-to-end decision and ranking pipelines for voice-based commerce, integrating user behavior modeling with real-time inference systems.
- Designed production systems for voice-enabled transactional flows with compliance constraints (e.g., parental consent, HIPAA-aligned systems).

Jun 2020 – May 2021 **Data Scientist / Software Developer,** ASU Decision Center for Educational Excellence.

- Developed predictive models for demographic, financial, and infrastructure planning using large-scale public datasets.
- Designed end-to-end data pipelines (AWS, ETL) for data ingestion, modeling, and deployment.
- Built models for resource allocation and policy analysis, identifying key drivers of graduation rates and community-level outcomes.

Academic Research Experience

2024 – Present **Mechanistic Interpretability of LLM Reasoning: Causal Mediation and Latent Interventions**

- Developed a causal mediation framework over intermediate reasoning tokens using activation patching and counterfactual inputs; showed that a sparse subset of tokens acts as stable causal mediators of the final answer, yielding an interventional account of where reasoning becomes causally committed inside the forward pass.
- Designed a training-free latent intervention method that identifies invariant reasoning directions via SVD of contrastive latent trajectories and constrains refinement to this subspace with an adaptive alignment gate; outperforms all latent refinement baselines on six reasoning benchmarks (+2.1% average), improves paraphrase consistency by $\sim$10%, and reduces latent trajectory variance by up to 50%, with zero parameter updates.

2025 – Present **Invariance-Constrained LLM Post-Training and Robust Evaluation**

- Developed a subspace-constrained preference optimization method that decomposes DPO updates into invariant task-relevant directions and unstable dataset-specific variation; deployed in production for catalog-scale product compatibility reasoning, improving incompatibility recall under paraphrase and distribution drift by 8% over standard DPO.
- Built causal-counterfactual evaluation protocols and trajectory-level stress tests that separate rule-following from pattern matching, measuring accuracy, calibration, and consistency across semantically equivalent inputs under perturbation and taxonomy drift.

2023 – 2025 **Causal Foundations: Invariant Structure Learning under Distribution Shift**

- Established the methodological basis for the agenda above: incremental and multi-view causal graph learning for non-stationary systems (IEEE TBD, CIKM, IEEE BigData), plus causally guided feature construction and privacy-aware transformation achieving $\sim$9% performance gain with $\sim$35% privacy leakage reduction.

Teaching, Mentorship & Academic Service

Invited Talks

Dec 2025 — **Keynote — IEEE International Conference on Modelling, Simulation & Intelligent Computing (MoSICom)**, BITS Pilani, UAE.
"Beyond Correlation: Can Causal Reasoning Enhance Modern AI Models in Robustness and Interpretability?"
Presented research on causal inference, robustness, and interpretability in modern AI systems, emphasizing limitations of correlation-based learning and the role of causal reasoning.

Teaching

Aug 2023 – Present — **Senior Teaching Assistant / Lab Instructor**, Arizona State University.

- Led instruction and lab sessions for courses in Object-Oriented Programming, Data Mining, and Semantic Web Mining for large undergraduate cohorts, and also project management for 30 Capstone Project teams.
- Designed and developed hands-on lab modules and course materials, emphasizing applied machine learning and data-driven reasoning.

Mentorship

Jan 2024 – Present — **Research Mentor**, Arizona State University.

- Mentoring undergraduate and graduate researchers on Neuro-symbolic AI for feature engineering, world model simulations, and LLM reasoning optimization.
- Guiding projects on investigating efficient fine-tuning and evaluation of large language models for reasoning tasks, including experimental design and model analysis.

Academic Service

Jan 2025 – Present — **Program Committee Member:** AAAI 2026; IEEE ICSCAN 2024; IEEE BigData 2024 (Undergraduate & High School Research Symposium); IEEE ICDM 2025.
Conference Reviewer: KDD, ICLR, NeurIPS, ICDM, IEEE BigData, Urban AI.
Journal Reviewer: ACM Transactions on Knowledge Discovery from Data (TKDD).
Session Chair: IEEE MoSICom 2025; IEEE ICDM 2025 — Undergraduate & High School Research Symposium.

Workshops & Outreach

Outreach

- Conducted hands-on workshops on Machine Learning fundamentals and Graph Neural Networks, training undergraduate students in applied AI techniques.
- Developed instructional material focused on practical model development and real-world applications.
- IEEE Phoenix Chapter Student Volunteer & Organizer

Honors & Awards

2025 — **IEEE ICDM Scholarship**, IEEE International Conference on Data Mining, 2025.
2018 — **Academic Scholarship**, Arizona State University, 2018.
2017–2018 — **Best Undergraduate Researcher Award**, 2017–2018.